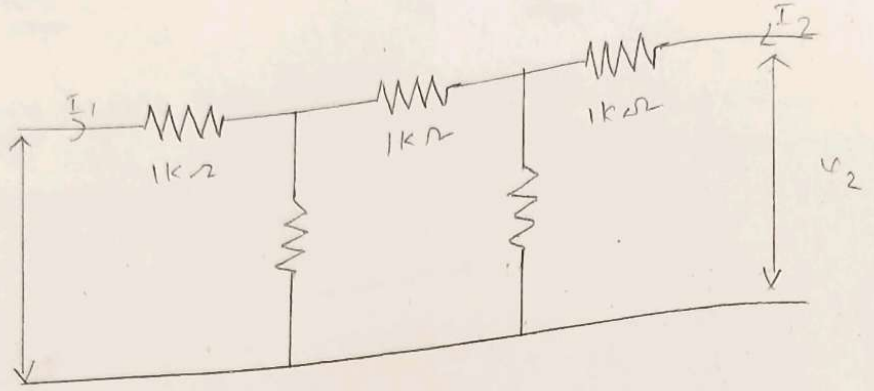
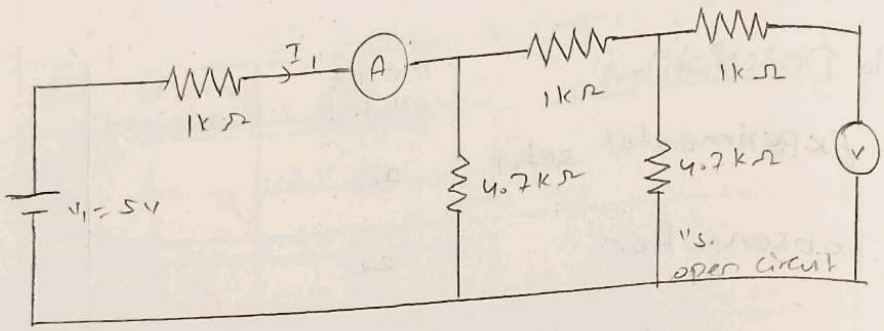
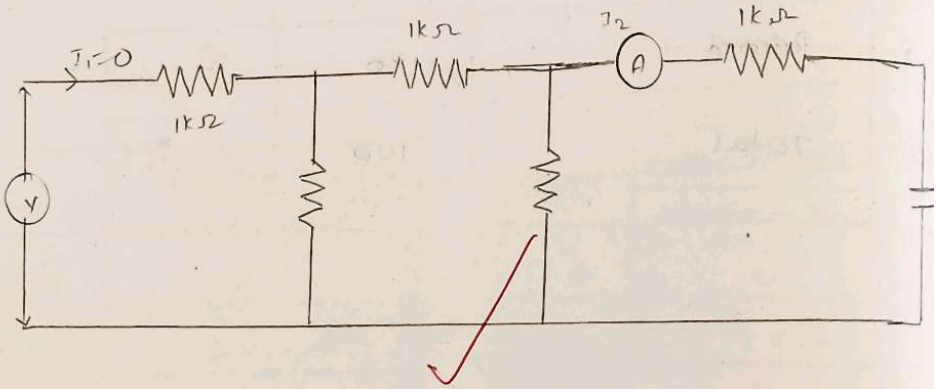
Circuit Diagram :-



Output open circuit :-



Input open circuit :-



Ex:07
Date:

DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORK.

Aim :-

To Calculate and verify Z Parameters of a two port network using multisim simulator.

Apparatus Required :-

S.No	Components	specification	quantity.
1.	Power supply	-	1
2.	multimeter	-	1
3.	Resistor	10k, 10k, 4.74	
4.	Bread board	-	1
5.	connecting wires	-	-
6.	pc with multisim simulator.	-	-

Tabulation:-

S.No	output open circuited $I_2 = 0$			$Z_{21} = V_2/I_1$ Ω	$Z_{11} = V_1/I_1$
	V_1	V_2	I_1		
1)	5	2	2	$Z_{21} = \frac{2}{2mA} = 1k\Omega$	$Z_{11} = \frac{5}{2mA} = 2.5$
2)	6	3	3	$Z_{21} = \frac{3}{3mA} = 1k\Omega$	$Z_{11} = \frac{6}{3} = 2mA$ $k\Omega$

S.No	output open circuited $I_1 = 0$			$Z_{12} = V_2/I_1$ Ω	$Z_{22} = V_1/I_1$
	V_1	V_2	I_2		
1)	5	2	2	$Z_{12} = \frac{5}{2} = 2.5k\Omega$	$Z_{22} = \frac{6}{8} = 2k\Omega$ $Z_{22} = \frac{5}{2} = 2.5k\Omega$
2)	6	3	3	$Z_{12} = \frac{3}{3} = 1k\Omega$	$Z_{22} = \frac{6}{3} = 2k\Omega$

PROCEDURE :-

1. connect the circuit as shown in fig and switch "ON" the experiment.

2. First open the o/p terminal and supply 5V to I/P terminal. Measure o/p voltage and I/P current.

3. Secondly, open I/P terminal and supply 5V to o/p terminal. Measure I/P voltage and o/p terminal current using multimeter.

4. Calculate the values of z parameter using equation 1 and 2.

5. Switch "OFF" the supply after taking the readings.

Calculation:-

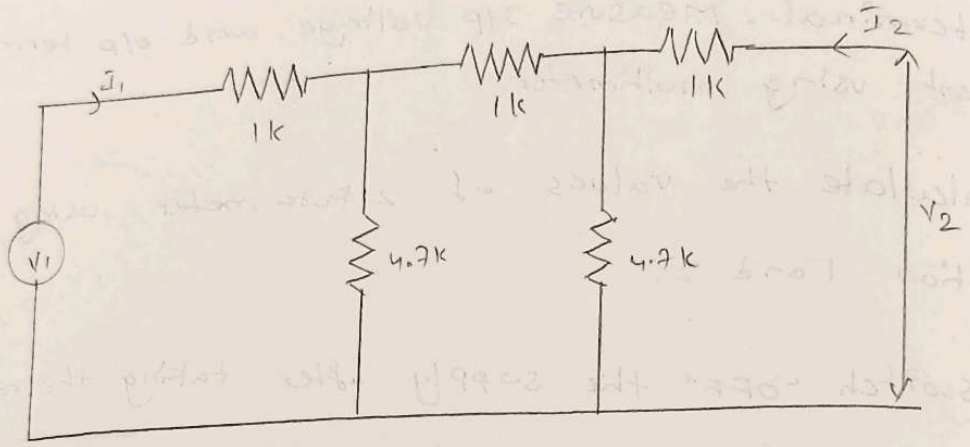
Theoritical calculation:

To Find Z_N and Z_{DL}.

$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

where $I_2 = 0 \Rightarrow$ Port 2 open circuit



$R_{eq} = 1 + 4.7 = 5.7$

$R_{eq2} = \frac{5.7 \times 4.7}{5.7 + 4.7}$

$= 2.575$

$R = 2.575 + 1 = 3.575\ k\Omega$

By ohm's law

$V_1 = I_1 R$

$\frac{V_1}{I_1} = R$

$Z_{11} = \frac{V_1}{I_1} = 3.575\ k\Omega$

By current Division Rule

$$I_2 = I_1 \times \frac{4.7}{4.7 + 4.7}$$

$$I_2 = \frac{4.7 I_1}{10.4}$$

$$I_2 = 0.451 I$$

ohm's law:

$$V_2 = I_2 \times 4.7$$

$$V_2 = 4.7 \times I_1 \times 0.451$$

$$\frac{V_2}{I_1} = 2.1197$$

$$Z_{21} = \frac{V_2}{I_1} = 2.1197 k\Omega$$

∴ the Network is symmetrical

$$Z_{11} = Z_{22}$$

$$Z_{21} = Z_{13}$$

$$Z = \begin{bmatrix} 3.575 & 2.1197 \\ 2.1197 & 3.575 \end{bmatrix}$$

S.No	DESCRIPTION	marks alloted	marks obtained
1.	Experimental setup	20	20
2.	observations	20	18
3.	calculations	20	20
4.	viva	20	18
5.	Record	20	18
6.	Total	100	96

Results :-

The 2-Parameter of two Parts network has been calculated and verified by using multisim simulator.